Newman-Penrose Formalism Calculations

Stephani et al., Equation (14.24)

Metric

$$ds^2 = -\frac{c^2 U_1\left(t,x\right)^2}{a^2 U\left(t,x\right)^2}\mathrm{d}t\otimes\mathrm{d}t + \frac{c^2}{U\left(t,x\right)^2}\mathrm{d}x\otimes\mathrm{d}x + \frac{c^2 e^{(-2\,x)}}{U\left(t,x\right)^2}\mathrm{d}y\otimes\mathrm{d}y + \frac{c^2 e^{(-2\,x)}}{U\left(t,x\right)^2}\mathrm{d}z\otimes\mathrm{d}z$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Metric Functions

$$U_1 = \frac{dU}{d(t+x)}$$

Null Tetrad

Covariant components

$$l = \left(\frac{\sqrt{2}cU_1\left(t,x\right)}{2\,aU\left(t,x\right)}\right)\,dt + \left(\frac{\sqrt{2}c}{2\,U\left(t,x\right)}\right)\,dx$$

$$n = \left(\frac{\sqrt{2}cU_1\left(t,x\right)}{2\,aU\left(t,x\right)}\right)\,dt + \left(-\frac{\sqrt{2}c}{2\,U\left(t,x\right)}\right)\,dx$$

$$m = \left(\frac{\sqrt{2}ce^{(-x)}}{2\,U\left(t,x\right)}\right)\,dy + \left(\frac{i\,\sqrt{2}ce^{(-x)}}{2\,U\left(t,x\right)}\right)\,dz$$

$$\overline{m} = \left(\frac{\sqrt{2}ce^{(-x)}}{2\,U\left(t,x\right)}\right)\,dy + \left(-\frac{i\,\sqrt{2}ce^{(-x)}}{2\,U\left(t,x\right)}\right)\,dz$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}aU\left(t,x\right)}{2\,cU_1\left(t,x\right)}\right)\,dt + \left(\frac{\sqrt{2}U\left(t,x\right)}{2\,c}\right)\,dx$$

$$n = \left(-\frac{\sqrt{2}aU\left(t,x\right)}{2\,cU_1\left(t,x\right)}\right)\,dt + \left(-\frac{\sqrt{2}U\left(t,x\right)}{2\,c}\right)\,dx$$

$$m = \left(\frac{\sqrt{2}U\left(t,x\right)e^x}{2\,c}\right)\,dy + \left(\frac{i\,\sqrt{2}U\left(t,x\right)e^x}{2\,c}\right)\,dz$$

$$\overline{m} = \left(\frac{\sqrt{2}U\left(t,x\right)e^x}{2\,c}\right)\,dy + \left(-\frac{i\,\sqrt{2}U\left(t,x\right)e^x}{2\,c}\right)\,dz$$

Directional Derivatives

$$D = -\frac{\sqrt{2}aU\left(t,x\right)}{2\,cU_1\left(t,x\right)}\,\partial_t + \frac{\sqrt{2}U\left(t,x\right)}{2\,c}\,\partial_x$$

$$\Delta = -\frac{\sqrt{2}aU\left(t,x\right)}{2\,cU_1\left(t,x\right)}\,\partial_t - \frac{\sqrt{2}U\left(t,x\right)}{2\,c}\,\partial_x$$

$$\delta = \frac{\sqrt{2}U\left(t,x\right)e^x}{2\,c}\,\partial_y + \frac{i\,\sqrt{2}U\left(t,x\right)e^x}{2\,c}\,\partial_z$$

$$\overline{\delta} = \frac{\sqrt{2}U\left(t,x\right)e^x}{2\,c}\,\partial_y - \frac{i\,\sqrt{2}U\left(t,x\right)e^x}{2\,c}\,\partial_z$$

Spin Coefficients

$$\rho = \frac{\sqrt{2}U\left(t,x\right)}{2\,c} - \frac{\sqrt{2}a\frac{\partial}{\partial t}U\left(t,x\right)}{2\,cU_{1}\left(t,x\right)} + \frac{\sqrt{2}\frac{\partial}{\partial x}U\left(t,x\right)}{2\,c}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{\sqrt{2}U\left(t,x\right)}{2\,c} + \frac{\sqrt{2}a\frac{\partial}{\partial t}U\left(t,x\right)}{2\,cU_{1}\left(t,x\right)} + \frac{\sqrt{2}\frac{\partial}{\partial x}U\left(t,x\right)}{2\,c}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = \frac{\sqrt{2}a\frac{\partial}{\partial t}U\left(t,x\right)}{4\,cU_{1}\left(t,x\right)} - \frac{\sqrt{2}\frac{\partial}{\partial x}U\left(t,x\right)}{4\,c} + \frac{\sqrt{2}U\left(t,x\right)\frac{\partial}{\partial x}U_{1}\left(t,x\right)}{4\,cU_{1}\left(t,x\right)}$$

$$\gamma = -\frac{\sqrt{2}a\frac{\partial}{\partial t}U\left(t,x\right)}{4\,cU_{1}\left(t,x\right)} - \frac{\sqrt{2}\frac{\partial}{\partial x}U\left(t,x\right)}{4\,c} + \frac{\sqrt{2}U\left(t,x\right)\frac{\partial}{\partial x}U_{1}\left(t,x\right)}{4\,cU_{1}\left(t,x\right)}$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{U(t,x)^2}{2c^2} + \frac{a^2 U(t,x)\frac{\partial^2}{(\partial t)^2}U(t,x)}{2c^2 U_1(t,x)^2} - \frac{aU(t,x)\frac{\partial^2}{\partial t\partial x}U(t,x)}{c^2 U_1(t,x)} + \frac{U(t,x)\frac{\partial^2}{(\partial x)^2}U(t,x)}{2c^2} - \frac{a^2 U(t,x)\frac{\partial}{\partial t}U(t,x)\frac{\partial}{\partial t}U_1(t,x)}{2c^2 U_1(t,x)^3}$$
$$- \frac{U(t,x)^2\frac{\partial}{\partial x}U_1(t,x)}{2c^2 U_1(t,x)} + \frac{aU(t,x)\frac{\partial}{\partial t}U(t,x)\frac{\partial}{\partial x}U_1(t,x)}{c^2 U_1(t,x)^2} - \frac{U(t,x)\frac{\partial}{\partial x}U(t,x)\frac{\partial}{\partial x}U_1(t,x)}{2c^2 U_1(t,x)}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{U(t,x)^2}{4c^2} + \frac{a^2 U(t,x)\frac{\partial^2}{(\partial t)^2}U(t,x)}{4c^2 U_1(t,x)^2} - \frac{U(t,x)\frac{\partial}{\partial x}U(t,x)}{2c^2} - \frac{U(t,x)\frac{\partial^2}{(\partial x)^2}U(t,x)}{4c^2} - \frac{a^2 U(t,x)\frac{\partial}{\partial t}U(t,x)\frac{\partial}{\partial t}U_1(t,x)}{4c^2 U_1(t,x)^3} - \frac{U(t,x)\frac{\partial}{\partial x}U(t,x)\frac{\partial}{\partial x}U_1(t,x)}{4c^2 U_1(t,x)} + \frac{U(t,x)^2\frac{\partial^2}{(\partial x)^2}U_1(t,x)}{4c^2 U_1(t,x)}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = -\frac{U(t,x)^2}{2c^2} + \frac{a^2 U(t,x)\frac{\partial^2}{(\partial t)^2}U(t,x)}{2c^2 U_1(t,x)^2} + \frac{aU(t,x)\frac{\partial^2}{\partial t\partial x}U(t,x)}{c^2 U_1(t,x)} + \frac{U(t,x)\frac{\partial^2}{(\partial x)^2}U(t,x)}{2c^2} - \frac{a^2 U(t,x)\frac{\partial}{\partial t}U(t,x)\frac{\partial}{\partial t}U_1(t,x)}{2c^2 U_1(t,x)^3}$$
$$- \frac{U(t,x)^2\frac{\partial}{\partial x}U_1(t,x)}{2c^2 U_1(t,x)} - \frac{aU(t,x)\frac{\partial}{\partial t}U(t,x)\frac{\partial}{\partial x}U_1(t,x)}{c^2 U_1(t,x)^2} - \frac{U(t,x)\frac{\partial}{\partial x}U(t,x)\frac{\partial}{\partial x}U_1(t,x)}{2c^2 U_1(t,x)}$$

$$\Lambda = -\frac{U(t,x)^2}{4c^2} + \frac{a^2\frac{\partial}{\partial t}U(t,x)^2}{2c^2 U_1(t,x)^2} - \frac{a^2 U(t,x)\frac{\partial^2}{(\partial t)^2}U(t,x)}{4c^2 U_1(t,x)^2} - \frac{U(t,x)\frac{\partial}{\partial x}U(t,x)}{2c^2} - \frac{\frac{\partial}{\partial x}U(t,x)^2}{2c^2} + \frac{U(t,x)\frac{\partial^2}{(\partial x)^2}U(t,x)}{4c^2}$$
$$+ \frac{a^2 U(t,x)\frac{\partial}{\partial t}U(t,x)\frac{\partial}{\partial t}U_1(t,x)}{4c^2 U_1(t,x)^3} + \frac{U(t,x)^2\frac{\partial}{\partial x}U_1(t,x)}{6c^2 U_1(t,x)} + \frac{U(t,x)\frac{\partial}{\partial x}U(t,x)\frac{\partial}{\partial x}U_1(t,x)}{4c^2 U_1(t,x)} - \frac{U(t,x)^2\frac{\partial^2}{(\partial x)^2}U_1(t,x)}{12c^2 U_1(t,x)}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{U(t,x)^2 \frac{\partial}{\partial x} U_1(t,x)}{6\, c^2 U_1(t,x)} + \frac{U(t,x)^2 \frac{\partial^2}{(\partial x)^2} U_1(t,x)}{6\, c^2 U_1(t,x)}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"